

Node anonymity in networks:

The infectiousness of uniqueness

Rachel de Jong, Mark van der Loo, Frank Takes

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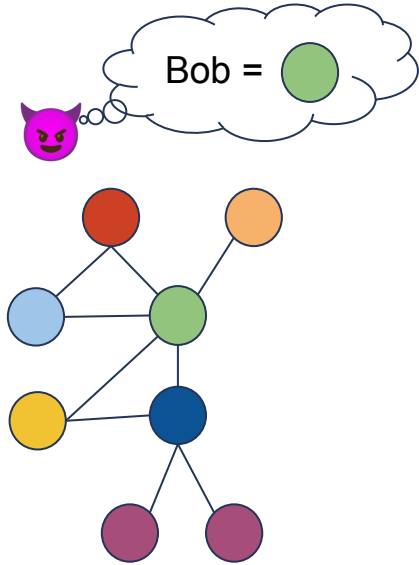
How anonymous are people in a network?

- Sensitive information
 - ☐ **Relations**
 - ☐ Attributes
- Comply with privacy laws
 - ☐ Identifiability of individuals
 - ☐ Information about individuals
- Trade off
 - ☐ Privacy vs. data utility

Goal: "unlocking network data"

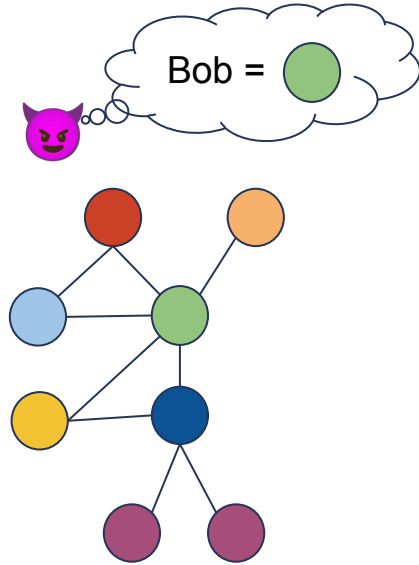


Anonymity in networks

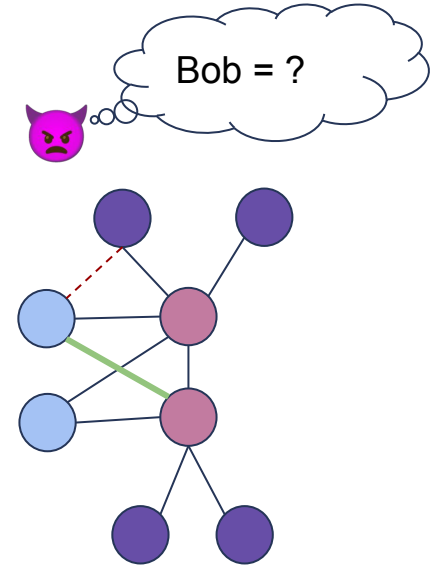


Original network

Anonymity in networks

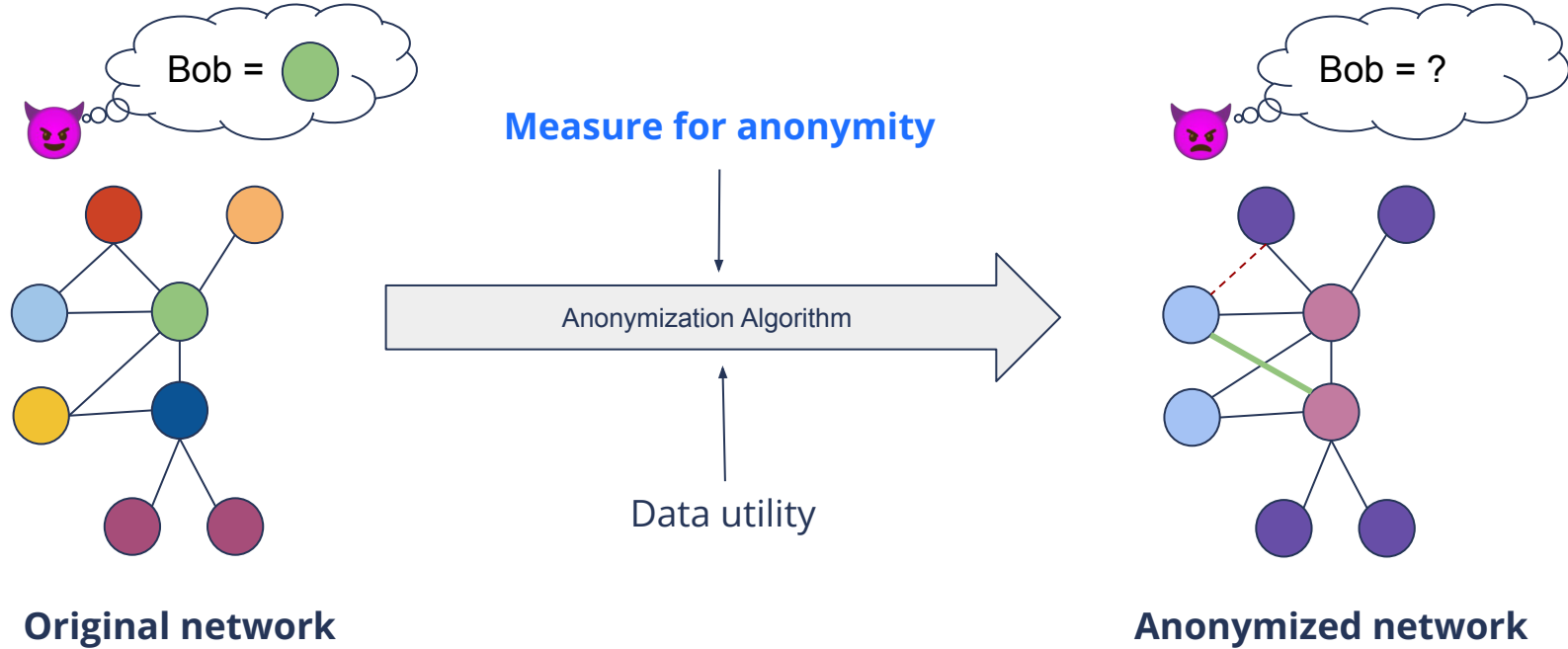


Original network



Anonymized network

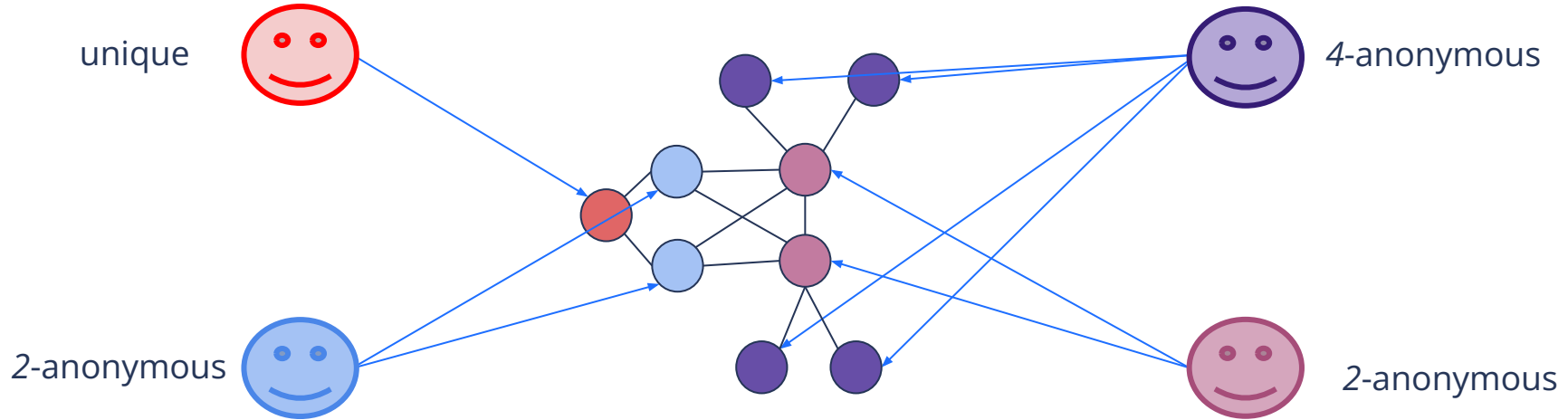
Anonymity in networks



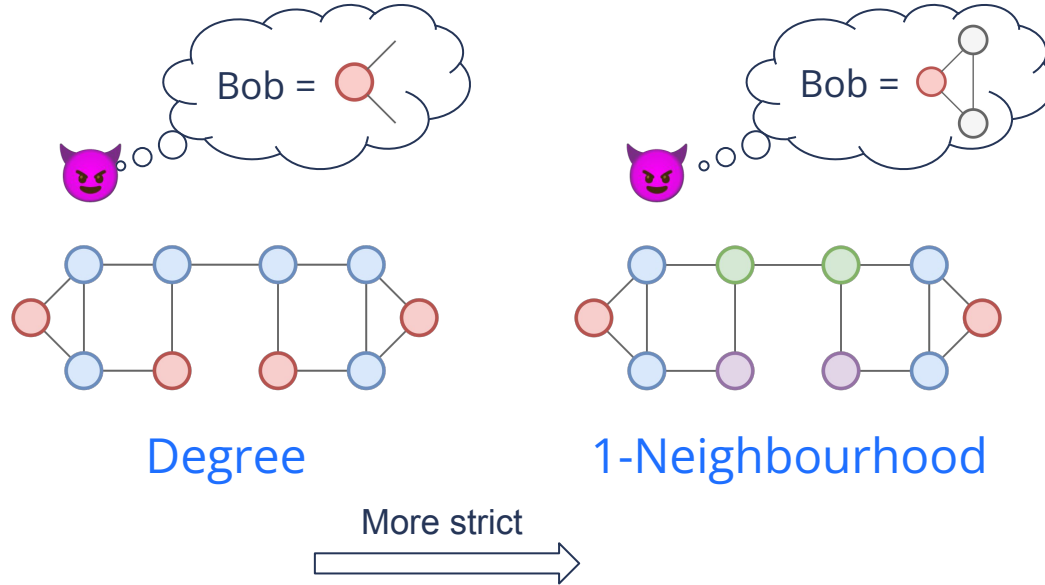
Definition: k -Anonymity

A node is k -anonymous if there are $k-1$ equivalent nodes

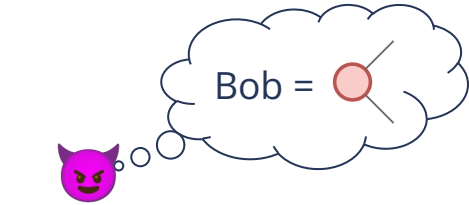
For now: anonymous = non-unique (2-anonymous)



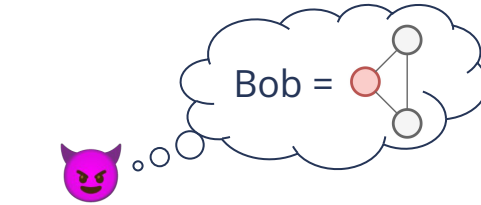
Measuring *network* anonymity



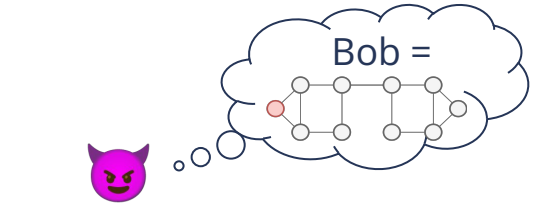
Measuring *network* anonymity



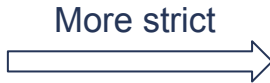
Degree



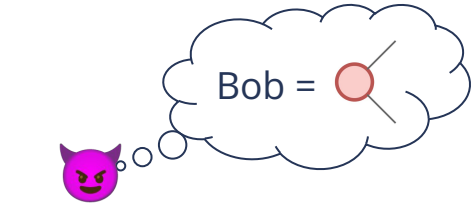
1-Neighbourhood



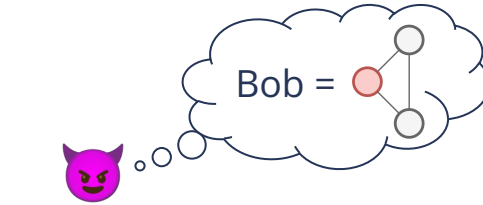
Automorphism



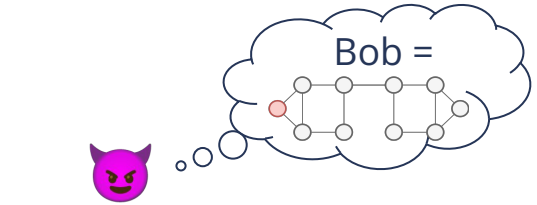
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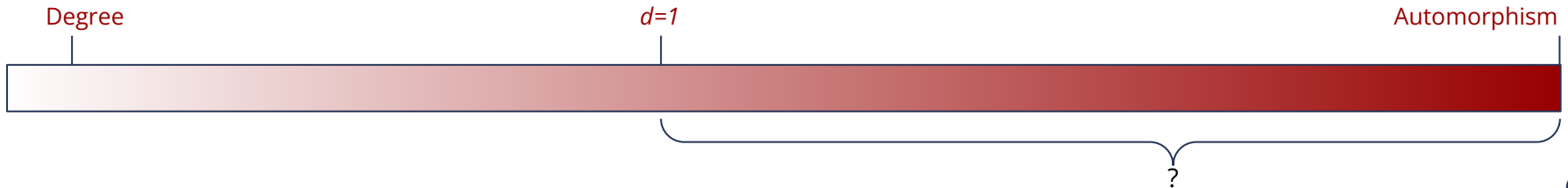
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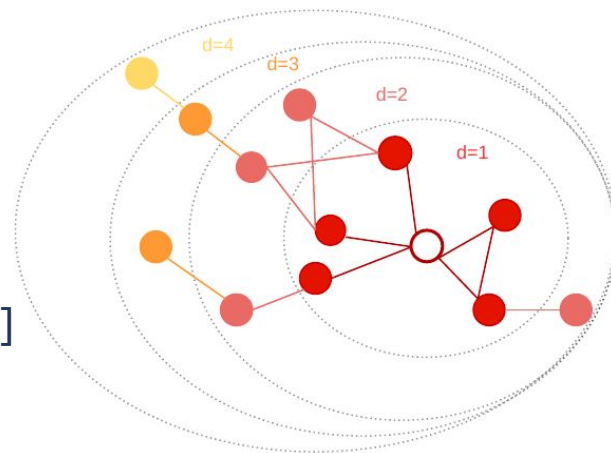


Definition: d - k -Anonymity

Nodes are equivalent if:

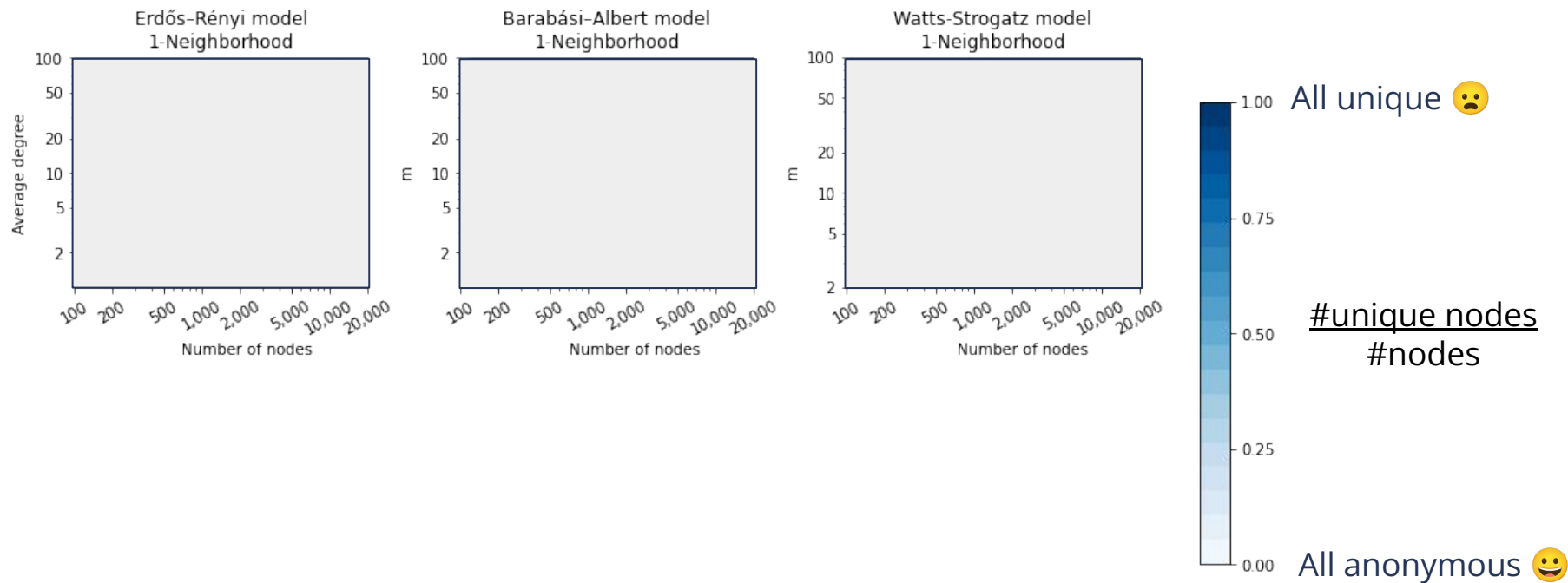
- d -Neighborhoods:
 - Are the same (*isomorphism*)
 - Nodes have the same structural position (*orbit*)

Computed using heuristics, iterative approach, etc. [1]

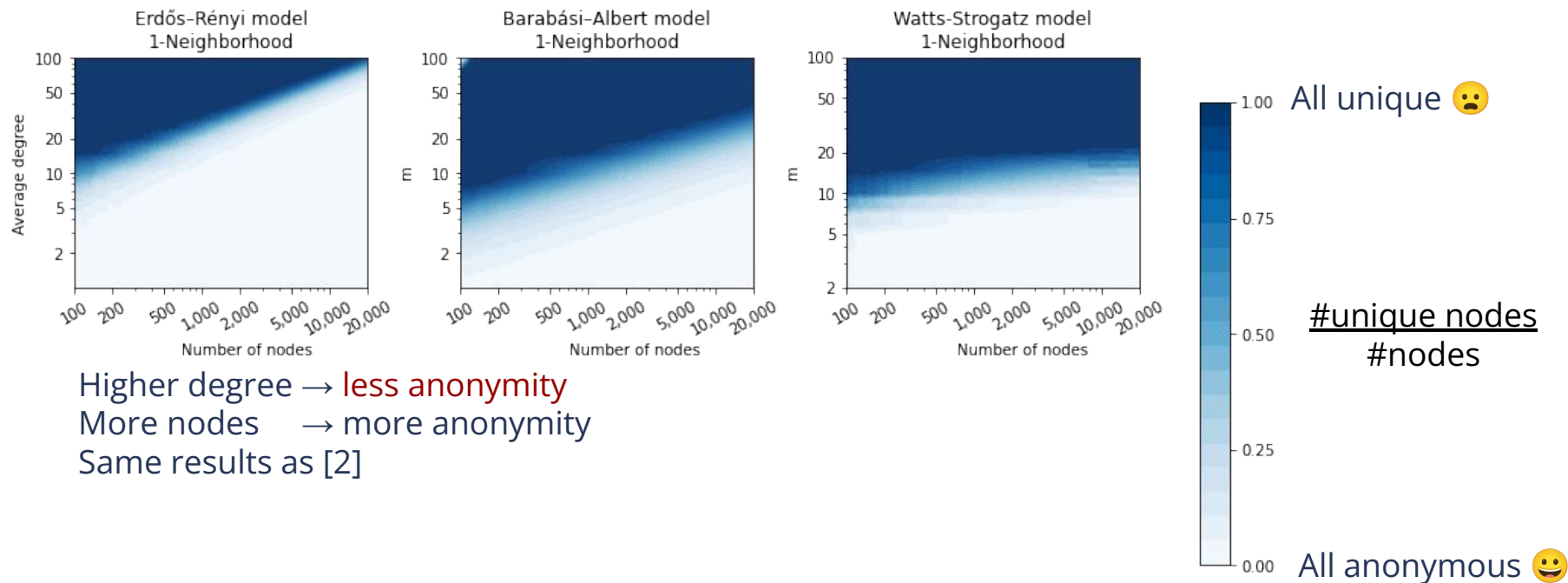


[1] de Jong, R. G., van der Loo, M. P. J. & Takes, F. W. Algorithms for efficiently computing structural anonymity in complex networks. ACM J. Exp. Algorithmics DOI: <https://doi.org/10.1145/3604908> (2023).

d - k -Anonymity on graph models

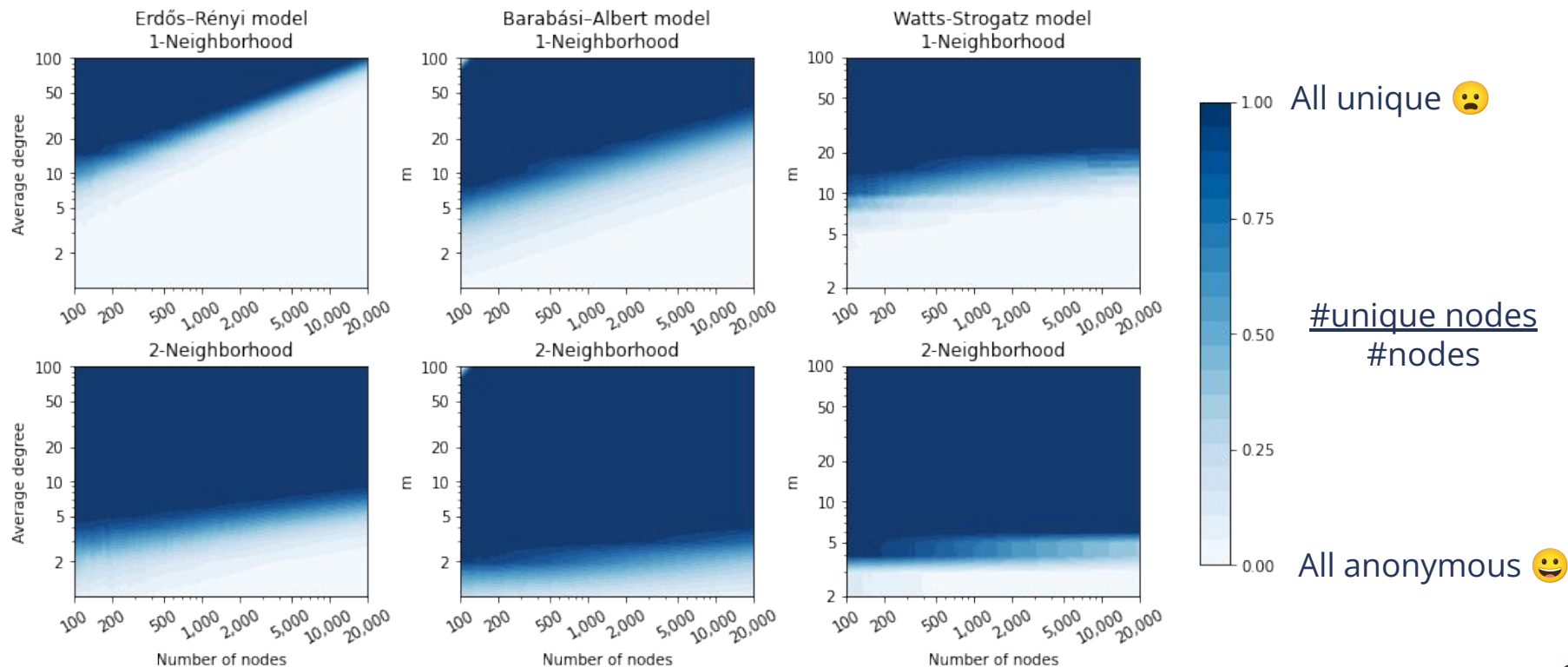


d - k -Anonymity on graph models



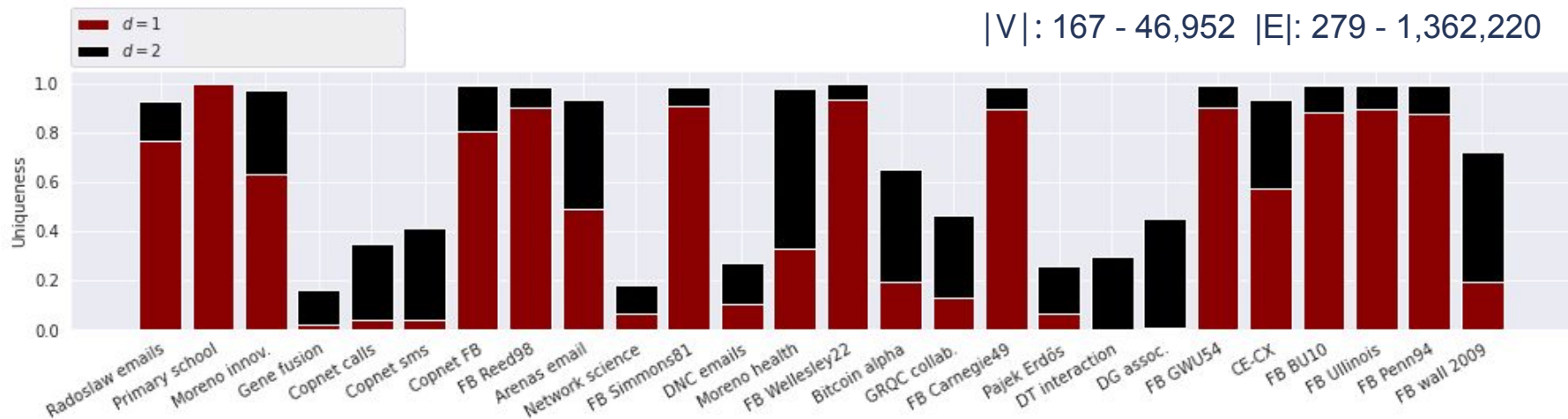
[2] Romanini, Daniele, Sune Lehmann, and Mikko Kivelä. "Privacy and uniqueness of neighborhoods in social networks." Scientific reports 11.1 (2021): 20104.

d - k -Anonymity on graph models

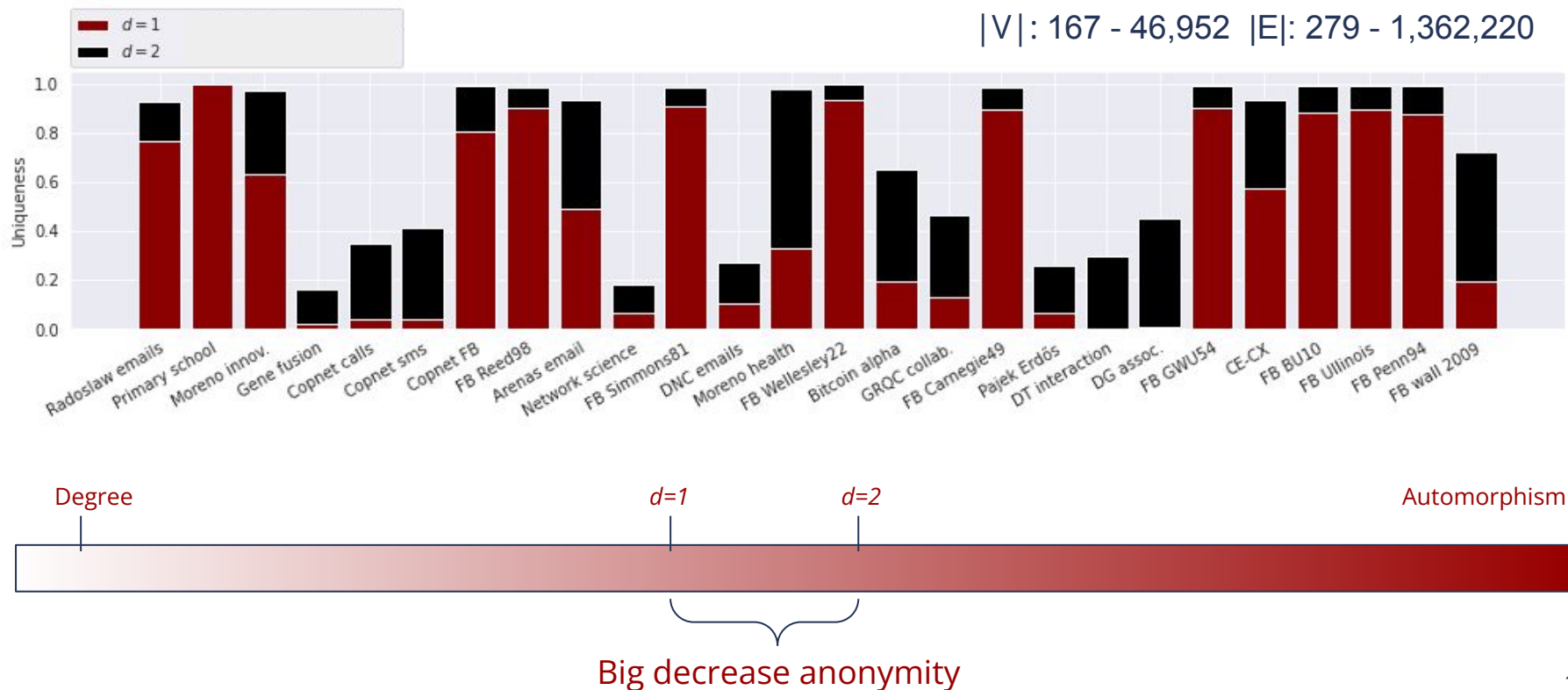


Almost no anonymity for $d=2$. Size has small / no effect

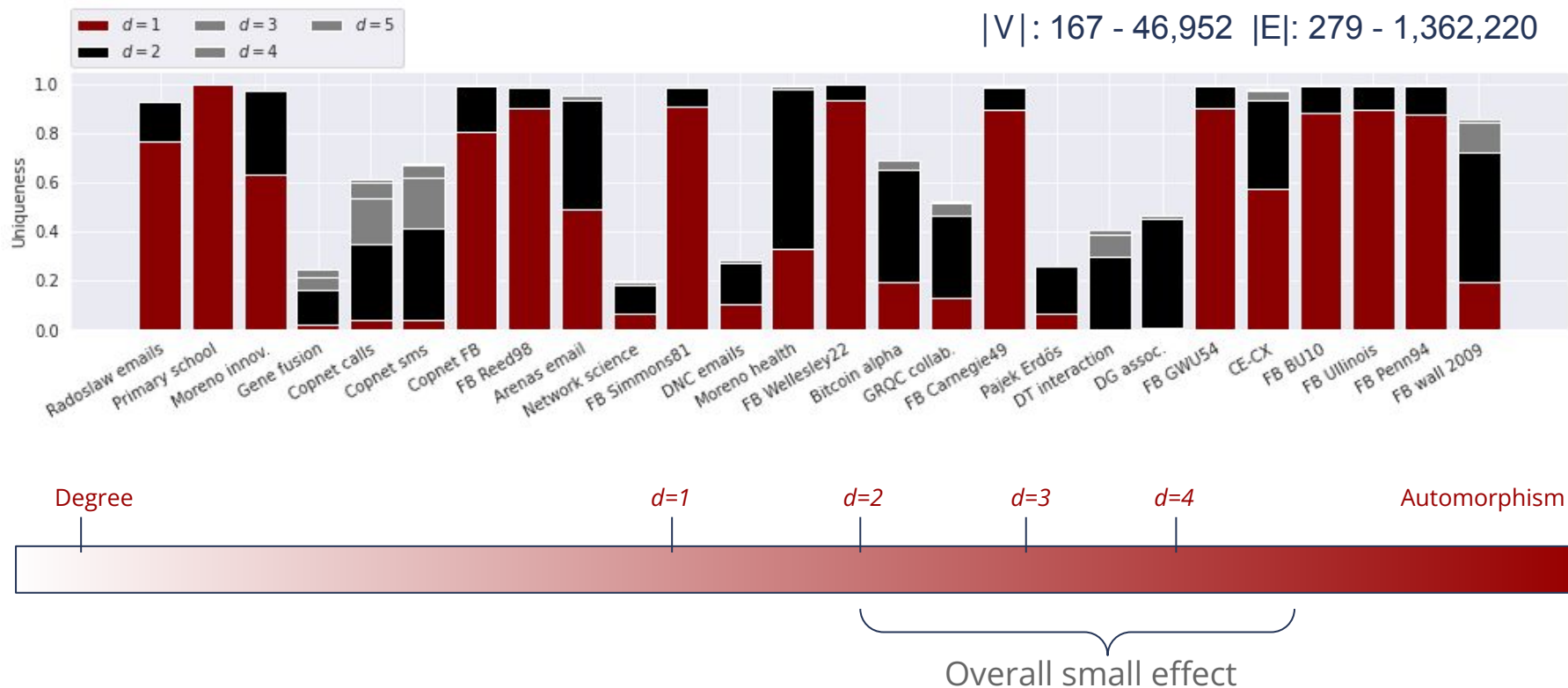
d - k -Anonymity on real networks



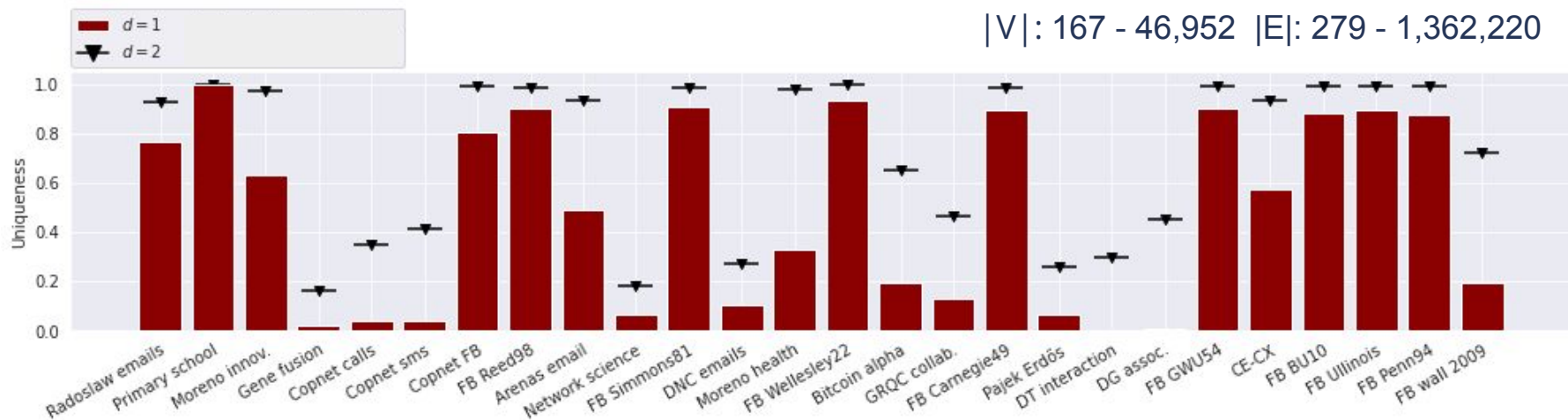
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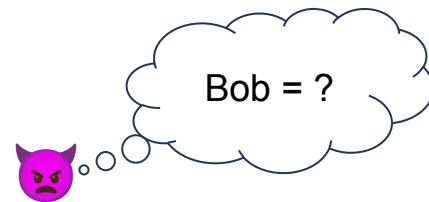


d - k - d - k -Anonymity on real networks



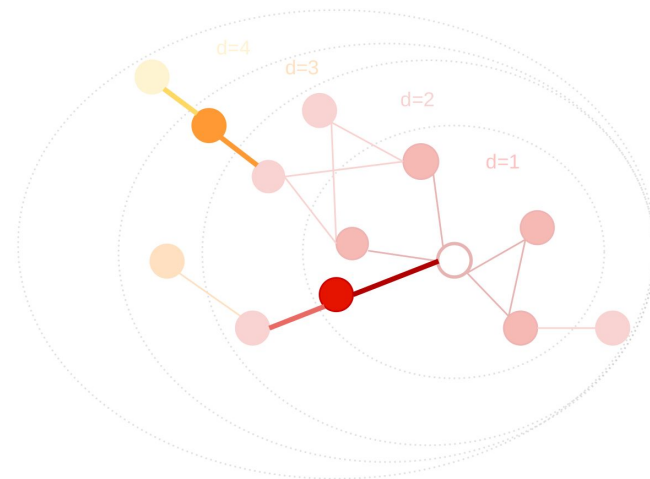
Big decrease anonymity

New approach: Anonymity-cascade

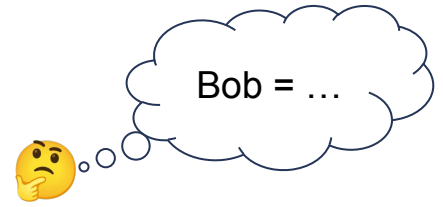


Knowledge:

- 1-neighborhood

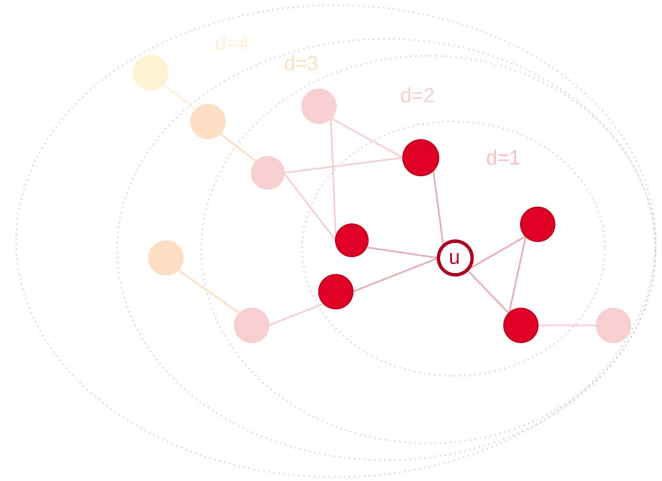


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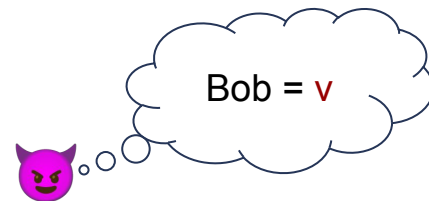


Knowledge:

- 1-neighborhood
- **Existence of a connection to unique node**

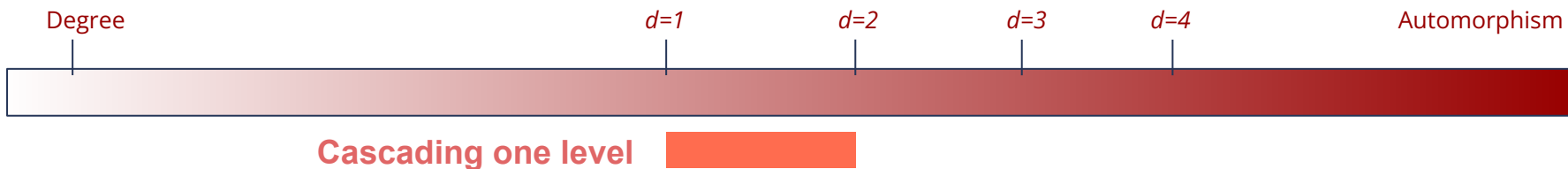
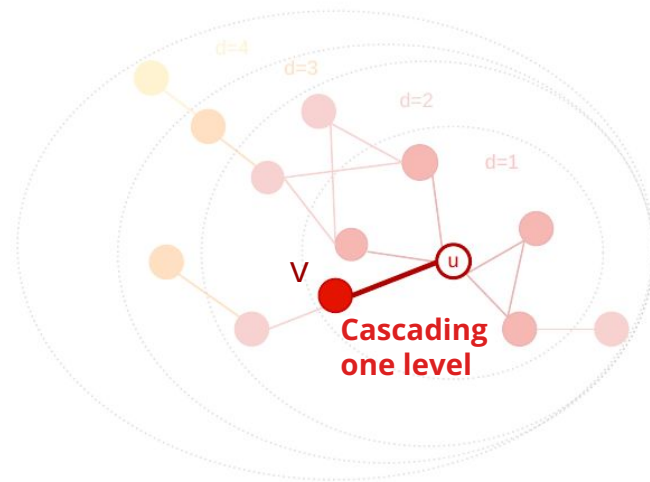


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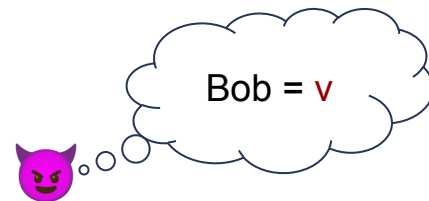


Knowledge:

- 1-neighborhood
- **Existence of a connection to unique node**
 - *"Infecting nodes with uniqueness"*
- Cascading one level is *at most as strict as* $d=2$

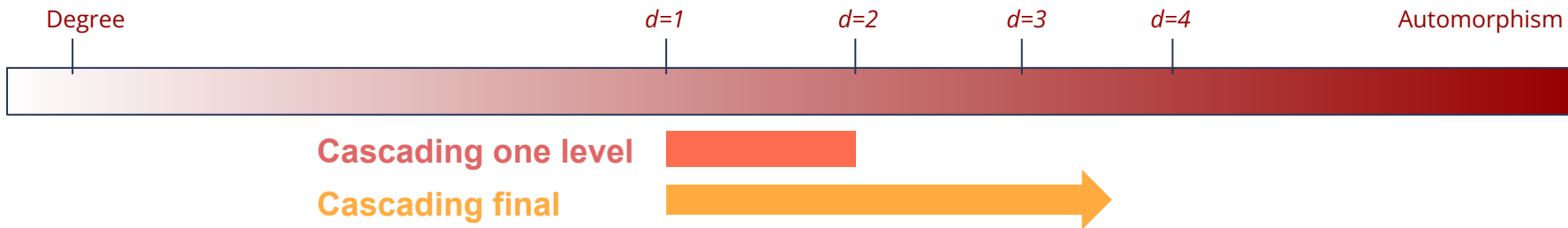
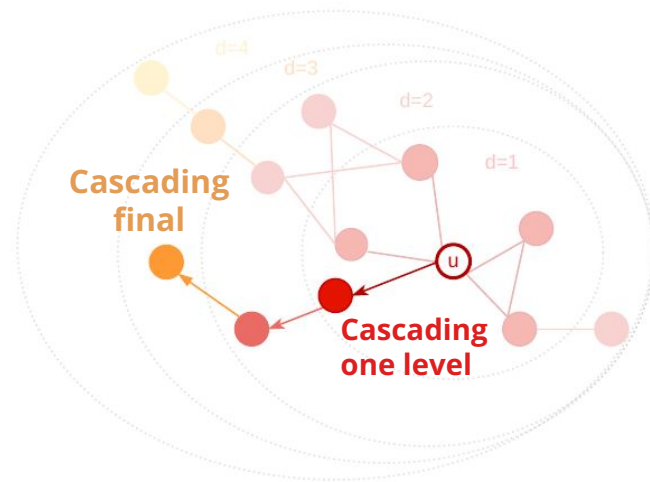


New approach: Anonymity-cascade

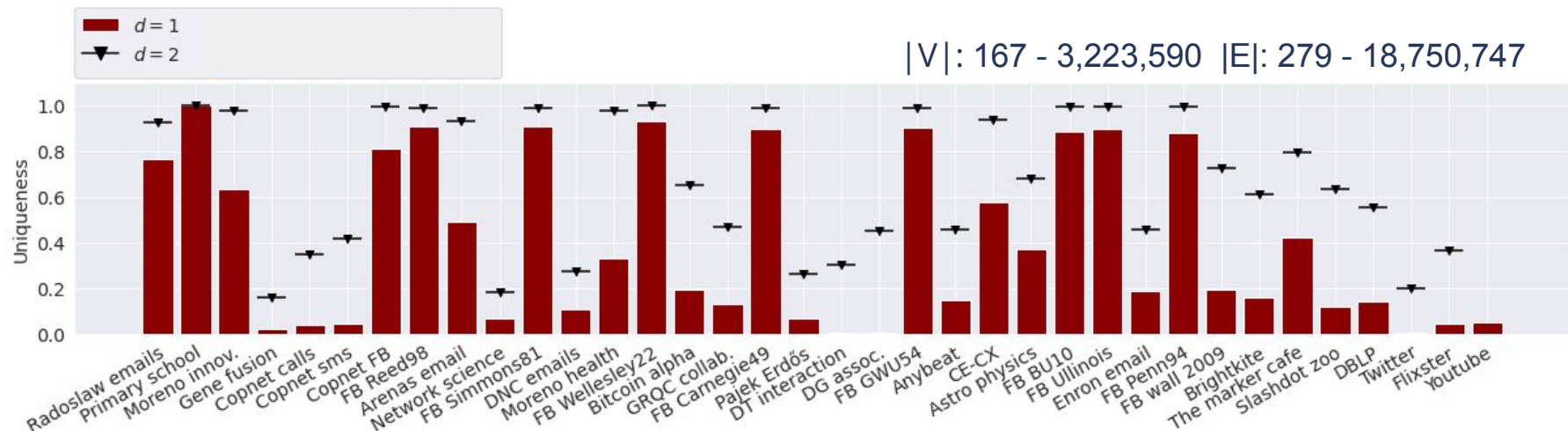


Knowledge:

- 1-neighborhood
- **Existence of a connection to unique node**
 - *"Infecting nodes with uniqueness"*
- Cascading one level is *at most as strict as $d=2$*
- Cascading final can be stricter

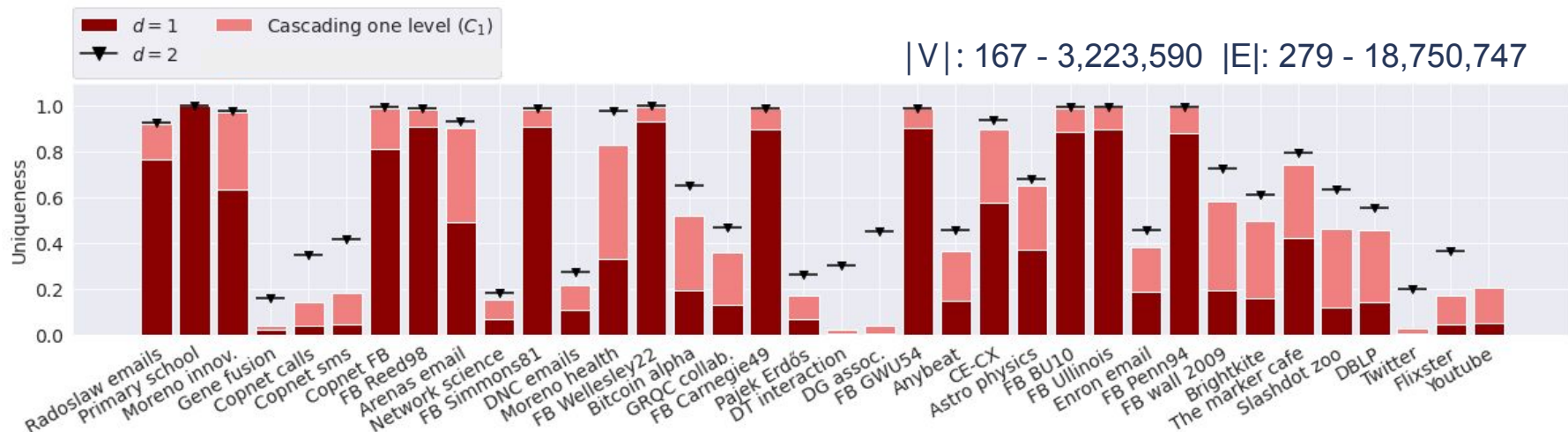


Anonymity-Cascade on real networks



*More networks due to lower runtimes $d=2$ and cascading

Anonymity-Cascade on real networks



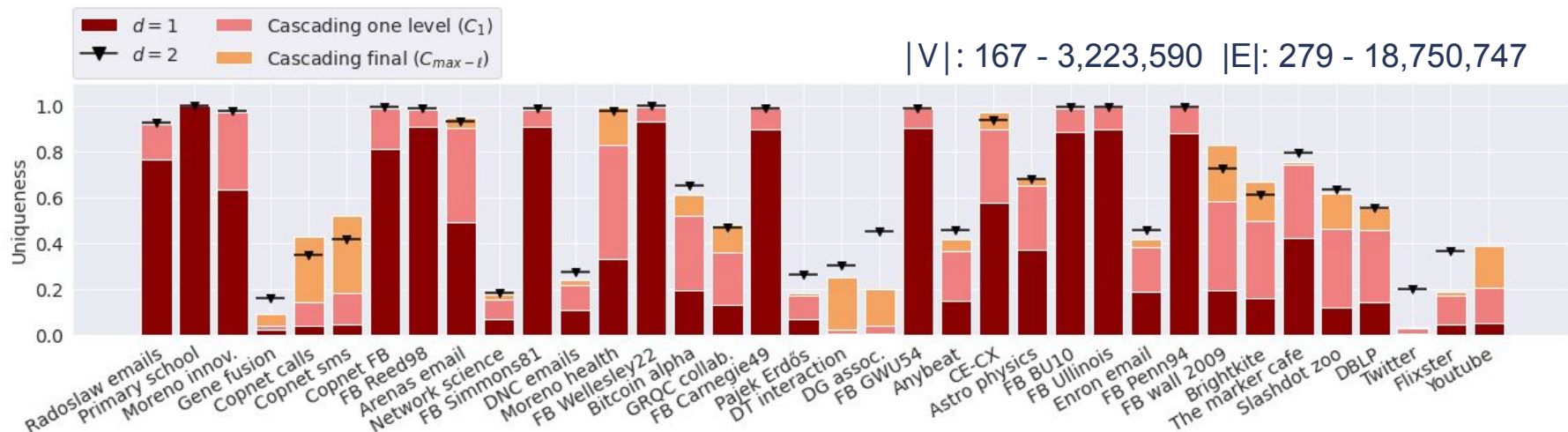
*More networks due to lower runtimes $d=2$ and cascading

>100% increase: **19** / 36

Large increase in uniqueness with knowledge of one connection

Approximate d - k -anonymity with $d=2$

Anonymity-Cascade on real networks



*More networks due to lower runtimes $d=2$ and cascading

>100% increase: 19 / 5 / 36

Large increase in uniqueness with knowledge of one connection

Approximate d - k -anonymity with $d=2$

Increase for some networks with cascading final

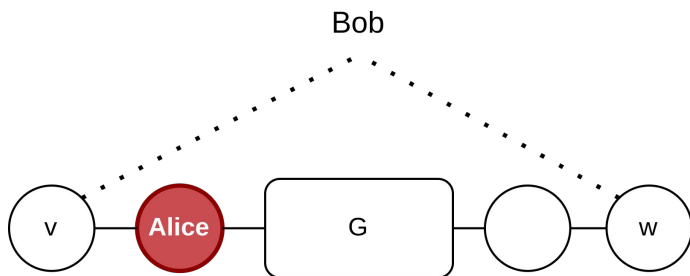
Network	Nodes	Edges	Fraction twins	$max - \ell$	Runtime		Runtime		Type
					d - k -anonymity $d=1$	$d=2$	Anonymity-cascade C_1	$C_{max-\ell}$	
Radoslaw emails ²⁸	167	3,250	0.072	3	0.02 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Primary school ²⁹	236	5,899	0.000	1	0.02 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Human contact
Moreno innov. ²⁸	241	923	0.025	3	< 0.01 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Gene fusion ²⁸	291	279	0.753	6	< 0.01 s	+ < 0.01 s	+ < 0.01 s	+ < 0.01 s	Biological
Copnet calls ³	536	621	0.287	10	< 0.01 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Copnet sms ³	568	697	0.285	7	< 0.01 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Copnet FB ³	800	6,418	0.005	4	0.02 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Online social
FB Reed98 ³⁰	962	18,812	0.012	3	0.05 s	+ 0.01 s	+ 0.01 s	+ 0.02 s	Online social
Arenas email ²⁸	1,133	5,451	0.042	5	0.02 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Communication
Network science ²⁸	1,461	2,742	0.755	6	0.01 s	+ 0.01 s	+ < 0.01 s	+ < 0.01 s	Co-autorship
FB Simmons81 ³⁰	1,518	32,988	0.011	3	0.12 s	+ 0.02 s	+ 0.01 s	+ 0.02 s	Online social
DNC emails ²⁸	1,893	4,385	0.706	4	0.02 s	+ 0.38 s	+ < 0.01 s	+ < 0.01 s	Online social
Moreno health ²⁸	2,539	10,455	0.003	5	0.03 s	+ 0.04 s	+ < 0.01 s	+ < 0.01 s	Human social
FB Wellesley22 ³⁰	2,970	94,899	0.000	4	0.4 s	+ 0.07 s	+ 0.04 s	+ 0.08 s	Online social
Bitcoin alpha ³⁰	3,783	14,124	0.306	5	0.04 s	+ 0.36 s	+ < 0.01 s	+ 0.01 s	Online social (trust)
GRQC collab. ³¹	5,242	14,484	0.455	8	0.03 s	+ 0.04 s	+ < 0.01 s	+ < 0.01 s	Co-autorship
FB Carnegie49 ³⁰	6,637	249,967	0.008	4	1.37 s	+ 0.44 s	+ 0.09 s	+ 0.18 s	Online social
Pajek Erdős ²⁸	6,927	11,850	0.737	6	0.02 s	+ 0.19 s	+ 0.01 s	+ 0.02 s	Co-autorship
DT interaction ³²	7,341	15,138	0.572	12	0.07 s	+ 2.88 s	+ < 0.01 s	+ < 0.01 s	Biological
DG assoc. ³²	7,813	21,357	0.531	8	0.21 s	+ 5.13 s	+ < 0.01 s	+ < 0.01 s	Biological
FB GWU54 ³⁰	12,193	469,528	0.006	4	2.64 s	+ 0.92 s	+ 0.18 s	+ 0.36 s	Online social
Anybeat ³⁰	12,645	49,132	0.500	5	0.41 s	+ 1.34 h	+ 0.02 s	+ 0.04 s	Online social
CE-CX ³⁰	15,229	245,952	0.021	6	1.04 s	+ 1.56 s	+ 0.09 s	+ 0.19 s	Biological
Astro Physics ³⁰	18,771	198,050	0.305	6	0.72 s	+ 1.91 s	+ 0.05 s	+ 0.11 s	Co-autorship
FB BU10 ³⁰	19,700	637,528	0.006	4	3.17 s	+ 1.44 s	+ 0.25 s	+ 0.50 s	Online social
FB Uillinois ³⁰	30,664	1,048,574	0.002	4	5.96 s	+ 2.66 s	+ 0.39 s	+ 0.78 s	Online social
Enron email ³¹	36,692	183,831	0.528	6	0.90 s	+ 1.44 m	+ 0.06 s	+ 0.12 s	Communication
FB Penn94 ³⁰	41,536	1,362,220	0.002	4	8.95 s	+ 5.39 s	+ 0.50 s	+ 1.00 s	Online social
FB wall 2009 ²⁸	46,952	183,412	0.133	8	0.54 s	+ 1.80 s	+ 0.05 s	+ 0.13 s	Communication
Brightkite ³⁰	58,228	214,078	0.258	8	0.78 s	+ 18.41 s	+ 0.06 s	+ 0.14 s	Online social
The marker cafe ³³	69,413	1,644,843	0.200	5	37.96 s	+ 10.46 m	+ 0.67 s	+ 1.35 s	Human contact
Slashdot zoo ²⁸	79,116	467,731	0.274	7	3.36 s	+ 2.62 m	+ 0.15 s	+ 0.34 s	Online social
Twitter ²⁸	465,017	833,540	0.801	5	59.53 s	+ 6.04 h	+ 0.29 s	+ 0.62 s	Online social
DBLP ²⁸	1,824,701	8,344,615	0.402	10	51.37 s	+ 11.58 m	+ 28.1 s	+ 57.18 s	Co-autorship
Flixster ²⁸	2,523,386	7,918,801	0.631	8	3.10 m	+ 8.57 h	+ 18.58 s	+ 37.81 s	Online social
Youtube ²⁸	3,223,585	9,375,374	0.000	15	13.24 m	+ > 1 week	+ 30.68 s	+ 1.04 m	Online social

Anonymity-cascade is
computationally efficient

Seconds vs.
minutes / hours!

Limitation: Twin Nodes

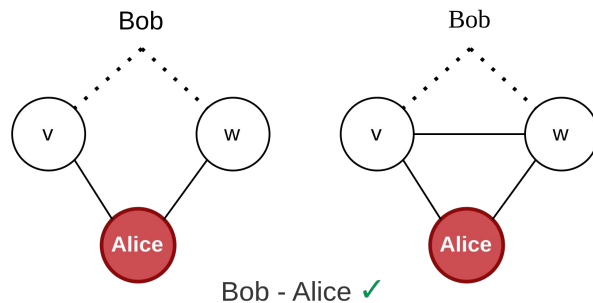
So far we have assumed that
k-candidates are distant



Bob - Alice ?

v and w structurally indistinguishable
(same orbit)

What if they are *both*
connected to Alice?



Open twins:

$$N_1(v) - \{v\} = N_1(w) - \{w\}$$

Closed twins:

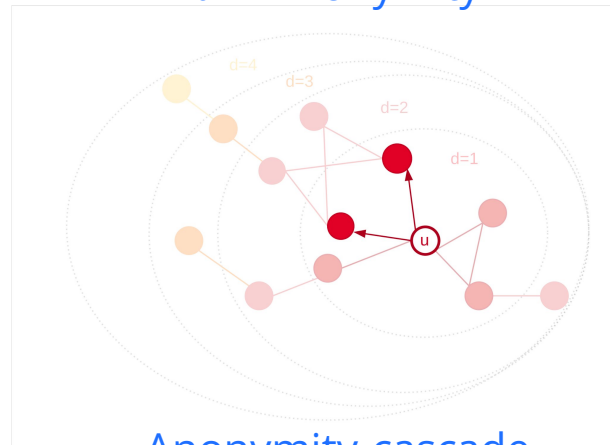
$$N_1(v) = N_1(w)$$

→ v, w structurally indistinguishable

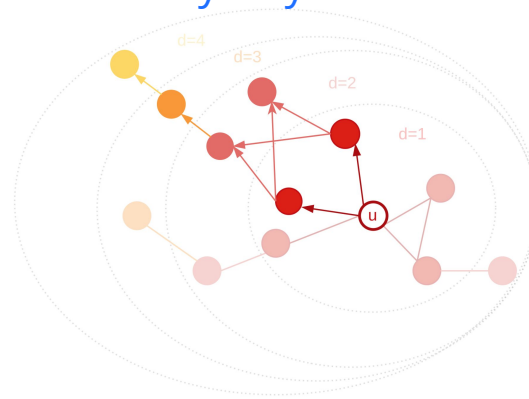
Limitation: Twin Nodes

- Twin nodes share their connections
 - **Structurally indistinguishable**
 - d - k -anonymity does not ensure privacy in the sense that have been considering
- Twin-unique:
 - Unique
 - All candidates are twins
 - d - k -anonymity + anonymity-cascade
 - Cascade from twin-unique nodes

d - k -Anonymity



Anonymity-cascade

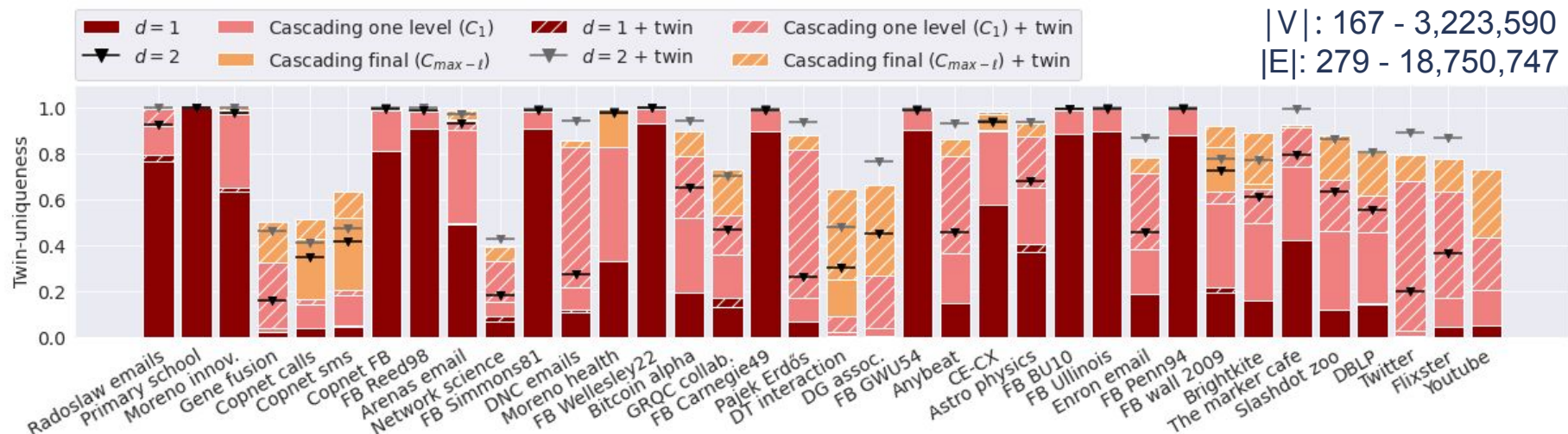


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Twitter ²⁸	465,017	833,540	0.801	5	59.53 s	+ 6.04 h	+ 0.29 s	+ 0.62 s	Online social
DBLP ²⁸	1,824,701	8,344,615	0.402	10	51.37 s	+ 11.58 m	+ 28.1 s	+ 57.18 s	Co-autorship
Flixster ²⁸	2,523,386	7,918,801	0.631	8	3.10 m	+ 8.57 h	+ 18.58 s	+ 37.81 s	Online social
Youtube ²⁸	3,223,585	9,375,374	0.000	15	13.24 m	+ > 1 week	+ 30.68 s	+ 1.04 m	Online social

Twin nodes can occur frequently: max **0.801**

Can not be unique!

Incorporating Twin-uniqueness



$C_1 \rightarrow$ Large increase twin-uniqueness compared to uniqueness

Conclusions

- To assess anonymity, we must look beyond the ego network
 - Substantial difference between $d=1$ and $d=2$
 - **d -k-Anonymity** looks beyond the direct neighborhood
- Uniqueness is infectious
 - **Anonymity-cascade** captures this effect
 - Knowledge of one extra link: large decreases in anonymity
- Twin nodes
 - Structurally indistinguishable, but occur frequently
 - Incorporating **twin-uniqueness** affects both d -k-anonymity and anonymity-cascade



Paper:

<https://arxiv.org/abs/2306.13508>

Thank you!

Rachel de Jong

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Leiden Computational Network Science group

🌐 cns.liacs.nl
🐦 @funwithnetworks

Other talks by our group at



Want to work with us in the Netherlands?
We are hiring! 3-year fully funded postdoc

- **Akrati Saxena:** Diverse Link Prediction in Complex Networks; July 11th at 14:40 PM
- **Carolina Mattsson:** Measuring the velocity of money; July 12th at 16:00 PM
- **Akrati Saxena:** Let's Tweet about Soccer? A Gender-centric Question; July 12th (Poster No. 62)
- **Akrati Saxena:** FairSNA: Algorithmic Fairness in Social Network Analysis; July 14th at 12:30 PM
- **Shane Mannion:** Fitting degree distributions of complex networks; July 14th at 14:30 PM
- **Rachel de Jong:** Node anonymity in networks: The infectiousness of uniqueness; July 14th 14:45 PM

Supplementary slides

Hackathon

Which information can be obtained?

May 3rd 2022 In 4 hours:

- 22 students found > 5,000 links about 26 volunteers
- Different sources: Facebook, linkedin, instagram (images)
- Different categories: Family, work, school
- 11 large pizzas

Co-organisers: Mark van der Loo, Jayshri Murli, Marieke de Vries, Peter-Paul de Wolf and, Frank Takes

HOW ANONYMOUS ARE YOU?!

How anonymous are you in openly available data? Is it possible for hackers to extract sensitive information from these data sets, by combining them with other data sources?

In this hackathon you, as a 'potential attacker', will search for network relations of a given list of people. If you are able to find the most relations within 4 hours, you receive a prize!

MAY 3, 15:00 - 19:00



SCAN TO REGISTER

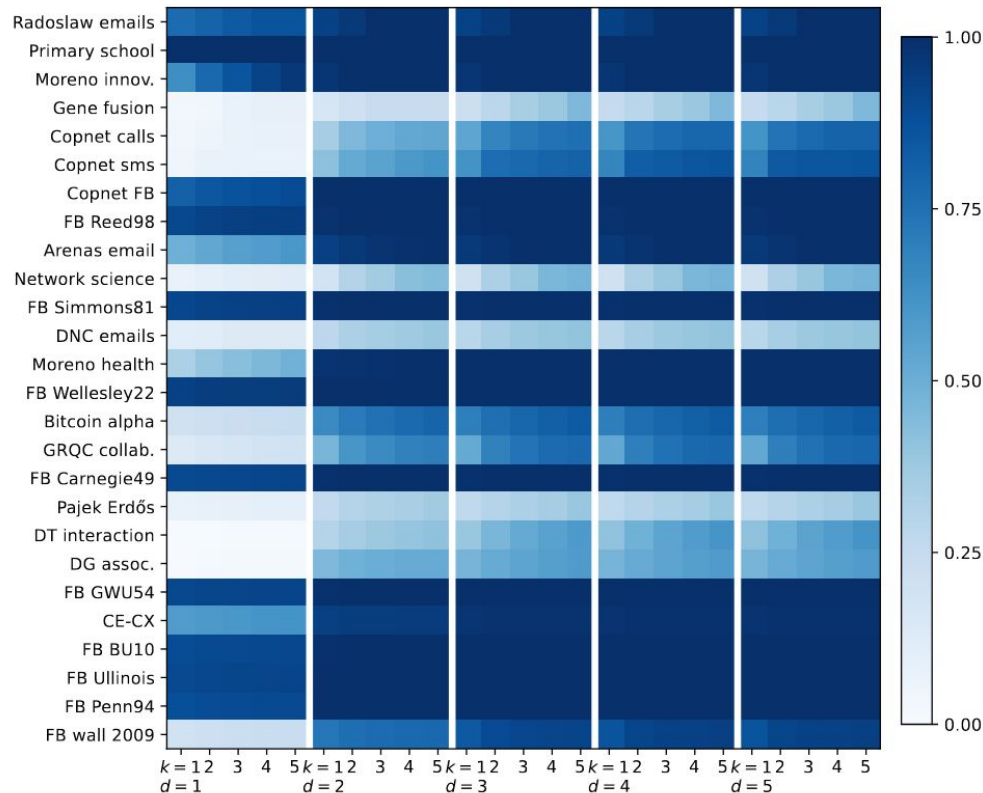
Free pizza and beverages afterwards!

Sign up before 27 April.



HACKATHON:
FROM PERSON TO
OPEN DATA

d - k -Anonymity: $d > 2$, $k > 1$



$d=1 \rightarrow d=5$

□ $d=1$ is not enough

□ After $d=2$ often not much change

$k=1 \rightarrow k=5$

□ Average: +0.05 up to 0.08

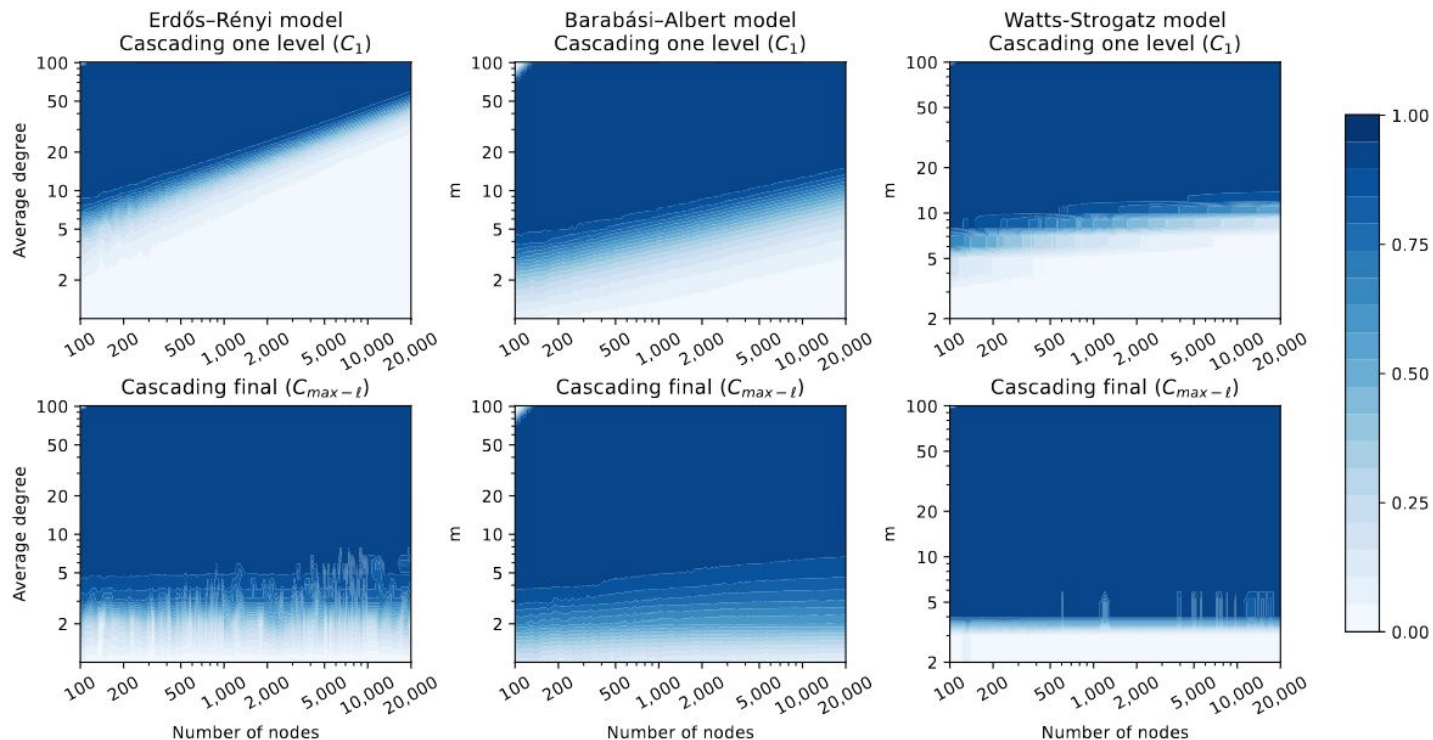
□ For some networks larger effect

Focus on:

□ 2-neighborhood

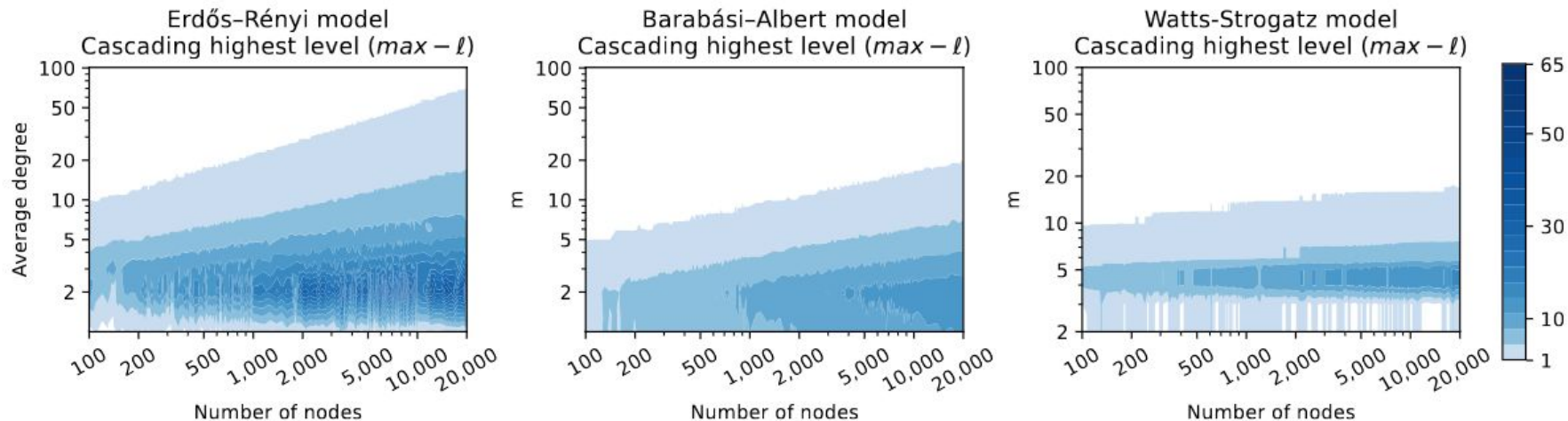
□ Uniqueness ($k=1$)

Anonymity-Cascade

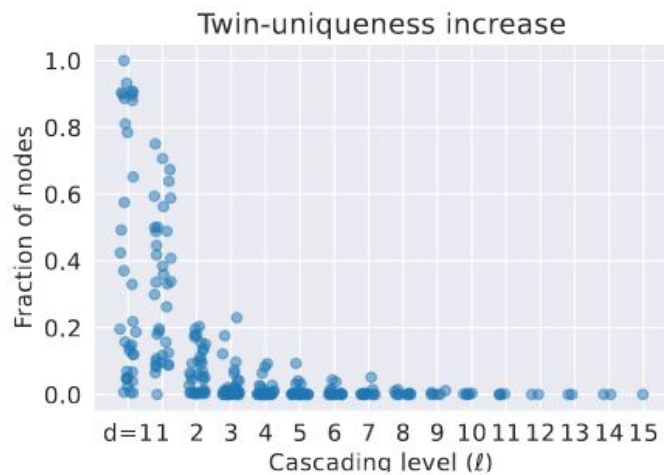
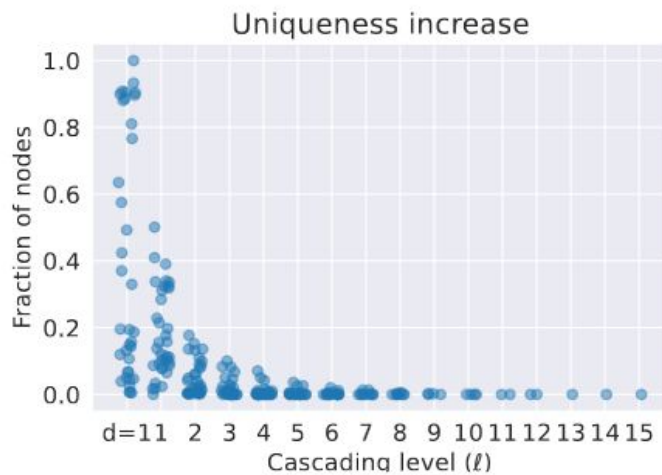


$\frac{\text{\#unique nodes}}{\text{\#nodes}}$

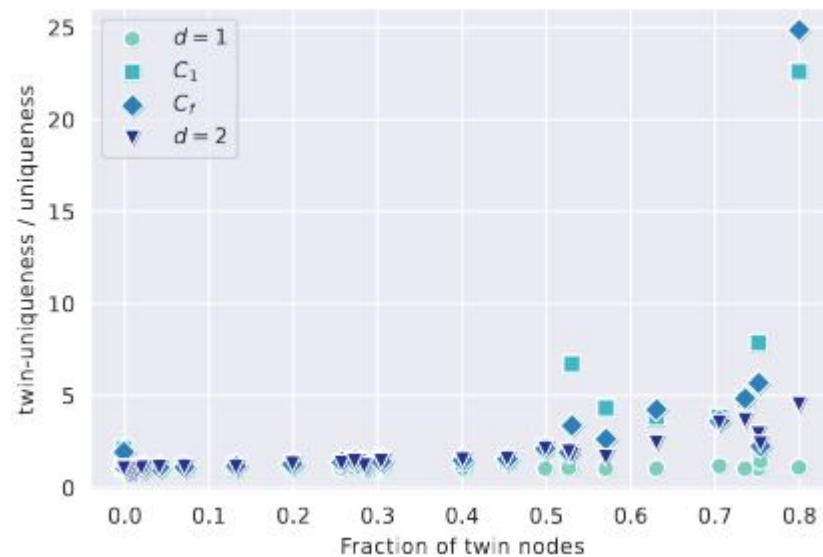
Anonymity cascade: levels



Cascading effect per level



Twin node effect



Measuring node anonymity in networks

1. d - k -Anonymity

2. Anonymity-cascade

3. Twin nodes

Q1: Is information about the ego-network enough?

Q2: Is plain neighborhood measurement sufficient?

Overview

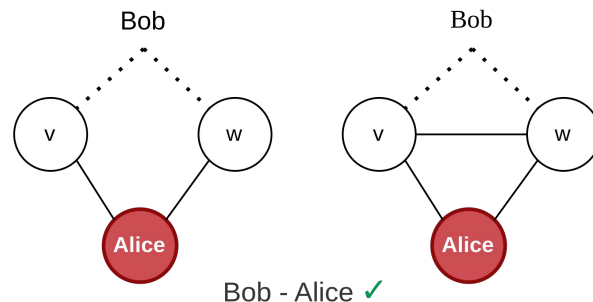
- What is anonymity in networks?
 - k -anonymity
- How to measure anonymity in networks?
 - d - k -Anonymity
 - Anonymity-cascade
 - Twin nodes
- How anonymous are people in a network?



Approach 3: Twin Nodes

■ Twin nodes share their connections

- Same degree
- Same 1-neighborhood
- Same 2-neighborhood
- ...
- Structurally indistinguishable!



Open twins:
 $N_1(v) - \{v\} = N_1(w) - \{w\}$

Closed twins:
 $N_1(v) = N_1(w)$

→ v, w structurally indistinguishable